WEEK 4 · LESSON 20 OF 24 · HANDOUT

Correlation: Do Two Things Move Together?

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=GtV-VYdNt_g — open the lesson online to watch inside the course.

🎓 **Go deeper:** freeCodeCamp: Statistics — a full university course — https://www.youtube.com/watch?v=xxpc-HPKN28

Your goal in this lesson: Measure how strongly two variables are related with covariance and the correlation coefficient, read a correlation matrix, and explain why correlation does not prove causation.

From a scatter plot to one number

A scatter plot shows whether two numbers move together. The **correlation coefficient r** puts it into one number from −1 to 1:

r	Meaning
close to **+1**	strong positive: when one goes up, the other goes up
close to **0**	no straight-line relationship
close to **−1**	strong negative: when one goes up, the other goes down

Rough guide: |r| above 0.7 is strong, 0.3–0.7 moderate, below 0.3 weak.

Run the setup cell (Kigali moto trips, simulated):

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
n = 300
distance = rng.uniform(1, 15, n).round(1)
rain = rng.random(n) < 0.25
hour = rng.integers(6, 23, n)
fare = (400 + 230 * distance + np.where(rain, 300, 0) + rng.normal(0, 150, n)).round(-1)
minutes = (distance * 2.8 + rng.normal(0, 3, n) + np.where(rain, 4, 0)).clip(3).round()
trips = pd.DataFrame({"distance_km": distance, "minutes": minutes, "fare_rwf": fare,
                      "hour": hour, "raining": rain})
```

Covariance and correlation

Covariance measures whether two variables move together, but its size depends on the units. **Correlation** divides covariance by both standard deviations, so it is always between −1 and 1 and has no units.

```
x = trips["distance_km"]
y = trips["fare_rwf"]
cov = ((x - x.mean()) * (y - y.mean())).sum() / (len(x) - 1)
r = cov / (x.std() * y.std())
print(round(cov, 1), round(r, 3))
print(round(np.corrcoef(x, y)[0, 1], 3))   # the same r, built in
```

Distance and fare have r close to 1: longer trips cost more, very consistently.

A correlation matrix

```
print(trips[["distance_km", "minutes", "fare_rwf", "hour"]].corr().round(2))
```

Each cell is the correlation between two columns. Analysts scan this table early in a project: features strongly correlated with the target are good candidates for a model. `hour` shows almost no correlation with fare — in this data, the time of day does not change the price.

r squared

r^2 is the share of the variation in one variable that a straight line on the other explains. If r = 0.98, then $r^2 \approx 0.96$: distance alone explains about 96% of the variation in fare. You will meet r^2 again when you evaluate regression models.

Correlation only sees straight lines

```
u = np.linspace(-3, 3, 100)
v = u ** 2
print(round(np.corrcoef(u, v)[0, 1], 3))   # about 0, yet v depends completely on u!
```

r near 0 does **not** mean "no relationship" — only "no straight-line relationship". Always look at the scatter plot too.

Correlation is not causation

Ice-cream sales and drowning accidents are positively correlated. Ice cream does not cause drowning — **hot weather** causes both. A hidden third variable like this is called a **confounder**.

Other traps: - **Reverse causation:** people who visit hospitals more are sicker — hospitals do not make them sick. - **Coincidence:** with enough variables, some will correlate by chance.

To show cause, you need careful experiments (like randomised trials) or strong reasoning about the mechanism. AI models learn correlations — so they can be fooled by confounders too.

Key terms

- **Covariance** — whether two variables move together (depends on units).
- **Correlation coefficient (r)** — strength and direction of a straight-line relationship, −1 to 1.
- **Correlation matrix** — a table of r for every pair of columns.
- **r^2** — share of variation explained by a straight-line relationship.
- **Confounder** — a hidden variable that affects both variables.
- **Causation** — one thing actually producing the other.

Remember this: Correlation measures how strongly two numbers move together in a straight line. Use it to find useful features — but always plot the data, and never treat correlation as proof of cause.

Name: _____ Date: _____

Activity

Task 1

What is the correlation between `distance_km` and `minutes`? Is it stronger or weaker than distance and fare?

Task 2

Calculate the correlation between `raining` (as 0/1: `trips["raining"].astype(int)`) and `fare_rwf`. What does it tell you?

Task 3

Think of a correlation from daily life in Rwanda that is probably caused by a confounder. Name the confounder.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What does r = -0.9 mean?
   A. No relationship
   B. A strong negative straight-line relationship
   C. A weak positive relationship
   D. An error
2. Why is correlation easier to compare than covariance?
   A. It uses more data
   B. It is always positive
   C. It has no units and is always between −1 and 1
   D. It is faster to type

3. $y = x^2$ for x from −3 to 3 gives $r \approx 0$. What does that show?

A. x and y are unrelated
B. Correlation only measures straight-line relationships
C. The code is wrong
D. r is always 0 for squares

4. Ice-cream sales and drowning are correlated. What is the likely explanation?

A. A confounder — hot weather increases both
B. Ice cream causes drowning
C. Drowning causes ice-cream sales
D. It is impossible

5. If r = 0.98 between distance and fare, about how much of the variation in fare does distance explain?

A. 49%
B. 2%
C. 98%
D. about 96%

6. Why do analysts look at a correlation matrix early in a project?

A. To delete data
B. To find features strongly related to the target
C. To draw maps
D. To prove causation

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